

are emerging as crucial players. These roles do not require complex programming; strong English and an understanding of prompting are often sufficient to begin working with large language models.

Looking ahead, tools like GitHub Copilot may become standard in embedded teams. We will see real-time large language models on edge devices, and companies like Nvidia or Intel may release their own hardware-specific models.

As the Internet of Things (IoT) continues to evolve, outdated devices pose a growing cyber threat. To address this, systems must be built with a secure architecture. That includes securing not only the devices themselves but also the platforms and servers they communicate with. End-to-end security from device to cloud can significantly reduce vulnerabilities.

There is growing interest in AI and IoT careers. While it may appear that fewer jobs are available, the reality is the opposite: opportunities exist for those who take proactive steps. Start by building a platform and testing it for scalability, security, and availability. Leading cloud providers such as AWS, Microsoft Azure, and Google Cloud offer microservices specifically designed for secure development. A hands-on approach can greatly enhance employability.

One innovative idea, suggested by a few EFY subscribers, involves simulating AI-driven interviews for specific roles, such as a former engineer at Synopsys in Bengaluru. In this concept, a candidate uploads their resume, and an AI engine generates interview questions one at a time using real-world data from platforms like Reddit, Quora, or Glassdoor. The candidate must answer each question before the next appears, mimicking real interview pressure and helping them prepare more effectively.

While such tools are promising, they must be used ethically and responsibly. There have been instances where candidates misuse AI during remote interviews by reading answers directly from a chatbot on-screen. This defeats the purpose of the exercise and undermines genuine skill-building.

It is now up to each of us to assess where we stand, how fast we want to grow, and how often we reflect—be it daily, weekly, or yearly.

So, focus on skills. Resources are everywhere. OpenAI, Google, and many others now offer free academies and courses—take advantage of them. Most importantly: Dream it. Believe it. Achieve it. Believe in your journey, take steady steps, and the results will follow. **EFY**

Based on a session titled, 'How to Build Your Global Career in IoT, Embedded Systems & AI' delivered by Abhijeet Desai, Group Project Manager, HCLTech America, at the EFY Expo held at the Auto Cluster Exhibition Center, Pune, on May 17, 2025. It has been transcribed and curated by Shubha Mitra, Assistant Editor at EFY.

Diotec 200W BLDC Motor Controller

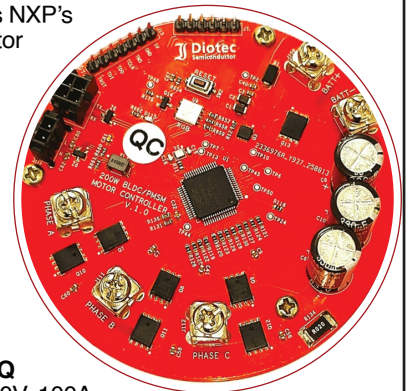
Reference Design Board – Automotive Electric Pump

The Diotec 200W BLDC Motor Controller is a compact, high-efficiency reference design board developed to demonstrate Diotec's power semiconductor portfolio in demanding motor control applications. Optimized for automotive pumps, fans and industrial drives, the design integrates Diotec MOSFETs, Schottky diodes and ESD protection devices to deliver a robust, thermally efficient and cost-optimized power stage.

Core Architecture

At the heart of the design is a 3-phase BLDC/PMSM inverter stage built with Diotec's high-performance MOSFET technology.

The board leverages NXP's S32M2 MCU for motor control algorithms, but the reliability, switching efficiency and protection features are driven by Diotec discrete components.



Key Features

Power Stage – DIOTEC

– DI110N04PQ-AQ

- Power MOSFET 40V, 100A, 2.5mΩ Rds(on), PowerQFN 5x6
- Ultra-low conduction losses
- Fast switching, low Qg for high efficiency
- 175°C operation for harsh environments
- Avalanche-rated for robust automotive reliability

Motor Control Features (Enabled by Diotec Power Stage)

- Field-Oriented Control (FOC) of BLDC/PMSM motors
- Sensorless and encoder-based operation
- Bi-directional rotation
- Single-shunt current sensing for reduced BOM and cost

Safety and Protection

- MOSFET based over current and short circuit protection
- DC-Bus over voltage and Under voltage safeguards
- Reliable start up fault detection

Target Applications

Automotive Systems

- Water/Oil Pump, HVAC Blower Motor, Cooling Fan
- Power Lift gate and seat control
- Electric Power Steering (EPS), Electronics Throttle Control (ETC)

FOR MORE INFO



Diotec Electronics India Private Limited

Survey No.614, Hissa No.3 to 9, Saswad Village, Purandar Taluka, Dist: Pune, Maharashtra - 412 301
Tel: +91-(0)-2115-225047/222523
Email: info@diotec.in; http://www.diotec.com/